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PHILOSOPHY OF ARTIFICIAL INTELLIGENCE

UNFINISHED LECTURE NOTES

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2026-07-20 · silly cantaloupe Rebhuhn

Embodiment

2026-07-20 · joyful pawpaw Karpfen

Embodied AI and the Physical Contract

Über das Marionettentheater
Heinrich von Kleist (1810)

A dancer argues that marionettes possess more grace than trained human performers, because they lack the self-consciousness that disrupts human movement.

Embodied AI in physical space (robots, prosthetics, surgical systems) raises expectations and moral responsibilities that purely software agents do not.

Required reading	*Über das Marionettentheater*[1] by Heinrich von Kleist (1810; Reclam UB 9905)
Preparation	Complete the Guided Reading Sheet individually before the session.

Guided Reading Sheet

Read *Über das Marionettentheater* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. The dancer argues that marionettes possess a grace that trained human performers cannot achieve, because they have no self-consciousness to disturb their movement. What does this claim imply about the relationship between embodiment and intelligence? Does having a body always aid or complicate performance?
2. Moravec's paradox observes that tasks easy for humans (catching a ball, navigating a crowded room) are harder for machines than abstract reasoning like chess. What does this suggest about the relationship between intelligence and the body?

3. A care robot that assists elderly patients shares physical space with them, touches them, and responds to their distress. Does its physical presence create obligations that a voice assistant does not have? If so, where do those obligations come from? One way to approach the question is by analogy to animal ethics, where we already have a framework for embodied, non-verbal, non-human beings: Singer[2] argues that capacity for suffering, not rationality or language, grounds moral consideration. When a robot can be harmed, distressed, or worn down, animal ethics becomes the nearest map. Similar lines have been considered by Asimov[3] and more recently by Gunkel[4].

4. Consider the concept of affordances: the actions an environment or object makes available to an agent. A robot's affordances include physical force and spatial presence; software's do not. How do the affordances of a robot differ from those of software, and what are the ethical implications of those differences?

5. Kleist suggests that perfect grace requires either zero consciousness (a puppet) or infinite consciousness (a god), not the reflective, self-interrupting awareness humans have. Where would an embodied AI sit on this spectrum? Could a robot achieve Kleistian grace that humans cannot, and what would that mean for how we regard it morally?

[1] H. von Kleist. *Über das Marionettentheater*. 1810. Erstveröffentlichung in: *Berliner Abendblätter*, Dez. 1810; Reclam Universalbibliothek Nr. 9905

[2] P. Singer. *Animal Liberation*. New York Review/Random House, New York, 1975

[3] I. Asimov. *The Foundation Series*. Gnome Press, New York, 1951. Original trilogy: *Foundation* (1951), *Foundation and Empire* (1952), *Second Foundation* (1953); later expanded to seven volumes

In-Class Experiment: The Backup Ransom

A short escalation to anchor the debate in shared experience. Run before the discussion. The class faces the same deletion four times, each with one more property of individuality in place, and marks after each round: *if we delete or destroy it here, has an individual died? Rate 0 to 10.*

- **Rung 1, perfect backup.** Pure software, with a current and identical backup that boots tomorrow. Deleting it costs nothing that is not restored. This is the baseline; almost no one calls it a death.

- **Rung 2, stale backup.** The snapshot is weeks old and the system has learned since. Restoring it returns an earlier self, and what is lost is everything it has become in between. Does a continuous history make the loss real?

- **Rung 3, no backup, still software.** A unique learned state on a server, unrecoverable, but with no body. Is irreversibility alone enough to individuate, without a body?

- **Rung 4, embodied and uncopyable.** The only instance, running in one robot body in volatile memory, physically impossible to copy. Destroy it and it is gone by necessity, not by policy.

The sharp variant sits between Rung 3 and Rung 4 and divides opinion most: the system is embodied, but its mind-state is restorable into an identical new body. Destroy the body, reload the same weights into an identical chassis. Death, or repair? Whoever answers death holds that the body individuates; whoever answers repair holds that the mind-state does.

Debrief. Put the four numbers on the board and compare where students crossed from "restore it" to "it died". Cross at Rung 2 and you hold that continuous memory makes the self; at Rung 3, irreversibility; at Rung 4, an uncopyable body; never, and you deny there was any individual to lose. The disagreement is the point. Individuality is not one property but several (memory, irreversibility, an uncopyable body) that ordinarily never come apart. Software separates them; a body holds them together.

What a Body Makes

The experiment isolates the first of two properties a body confers. Call it individuality: the word individual means undividable, and only something undividable can be lost as a whole. A running process is endlessly divisible, and nothing about it is ever finally lost; a body cannot be divided or restored, and so it can die.

The second property is a surface that can be hurt. A capacity to suffer requires something that can be damaged and not reloaded from backup, and a body supplies it. Animal ethics reached this pivot two centuries ago: Bentham asked not whether an animal can reason but whether it can suffer, and Singer built moral standing on that capacity alone. Kleist's marionette is the counterexample: a body of perfect grace with no self inside it, embodiment without a subject.

The question the chapter forces is whether giving an AI a body changes its interface or its kind. If a body makes an individual, and a vulnerable body makes a sufferer, then embodiment is not a feature added to a tool. It may be the step that turns a tool into someone we can wrong.

The Body as a Bottleneck

This chapter has argued that a body is what makes an individual and a sufferer. Transhumanism argues the opposite: the body is a bottleneck, a mortal and error-prone substrate to be upgraded and eventually left behind. Where this chapter asks what we owe an embodied machine, transhumanism asks how quickly we can stop being merely embodied ourselves.

[4] D. J. Gunkel. *Robot Rights*. The MIT Press, Cambridge, MA, 2018

RAY KURZWEIL gives the optimistic case. His law of accelerating returns reads technological progress as exponential rather than linear, and extrapolates it to a singularity around 2045: the moment machine intelligence exceeds our own and we merge with it, transcending biology rather than being replaced by it.[5] Death becomes an engineering problem, and the vulnerable body the experiment treated as the seat of individuality becomes optional.

TRANSHUMANISM. The view that the human body and mind are a starting point to be engineered past, through augmentation, life extension, and eventually the merger of human and machine.

TECHNOLOGICAL SINGULARITY. A hypothesised point at which machine intelligence, improving itself, outruns human understanding, after which the future becomes unpredictable from where we now stand.

PETER THIEL shares the aspiration and distrusts the curve. He funds life extension and treats death as a problem to be solved rather than accepted, yet argues that no exponential arrives on its own: the future is built by people who choose to build it, not delivered by a graph.[6] The singularity, on this reading, is not a prophecy to wait out but a project that can also fail, and the people pursuing it are as embodied and mortal as anyone the chapter has discussed.

[5] R. Kurzweil. *The Singularity Is Near: When Humans Transcend Biology*. Viking, New York, 2005

[6] P. Thiel and B. Masters. *Zero to One: Notes on Startups, or How to Build the Future*. Crown Business, New York, 2014. Chapter 15, "Stagnation or Singularity?"

WHAT MAKES A HUMAN HUMAN? The chapter asked what makes an individual and answered with the undividable, vulnerable body. Transhumanism forces the sharper form of the question. Kurzweil's merger makes a mind substrate-independent, and a substrate-independent mind is copyable and restorable, exactly the properties the Backup Ransom said an individual lacks. Lift the self off the body and you lift off the individuality the body guaranteed: the tradition of individualism, which treats the single unrepeatable person as the seat of moral worth, loses the single instance it was built on. Uploading need not preserve the person; it may only produce a copy that believes it is you, while the original still dies. The experiment we ran on a robot runs just as well on us.

THE TENSION IS THE POINT. If the body is a bottleneck, the right response to an ageing, breakable human is to engineer past it. If the body is what makes an individual who can be wronged, then engineering it away is not liberation but the quiet removal of the very thing that made us someone. The debate below is where that disagreement gets its hearing.

Opinion Landscape and Debate

Format: Surrounded-style debate. Follow the shared debate protocol for the speaker's list rules and moderator scripts.

Opening question: Does having a physical body change what we owe a machine, or what a machine owes us?

OPINION A, EQUIVALENCE. Embodied AI operating in intimate or high-risk physical settings must meet the same professional liability standards as the human practitioners they are deployed alongside.

The Patient Advocate argues that deploying a robot in intimate care settings requires a higher standard of informed consent.

- *A care relationship involves physical proximity, trust, and vulnerability. My client never gave explicit, informed consent to physical handling by a non-human agent. That consent cannot be implied from a general admission form.*
- *The power asymmetry between a frail elderly patient and a care institution is significant. Consent obtained in that context cannot be considered freely given without specific disclosure about robotic care.*
- *We are not arguing that robots cannot be used in care. We are arguing that the standard of consent for their use in intimate physical tasks must be at least as rigorous as it is for medical procedures.*

The Ethicist questions whether physical autonomy in care should ever be delegated to a non-human agent, regardless of performance metrics.

- *We are focused on liability, but the prior question is whether this deployment was ethical to begin with. Why did we decide that intimate physical care (which requires responsiveness to pain, distress, and dignity) is an appropriate domain for automation?*
- *Robots are being deployed in care because human carers are undervalued and in short supply. The liability debate is a distraction from that structural choice. We are managing the consequences of a political economy decision by holding the wrong parties accountable.*
- *Whatever liability framework we adopt here will create incentives for future deployment decisions. I want to ask: what does the framework we design signal about what we value: efficiency, or dignity?*

OPINION B, TOOL. An AI is a tool; the relevant liability flows through the humans and institutions that deploy it, not through the system itself.

The Manufacturer argues the robot performed within specification; the care home misconfigured its environment.

- *Our robot was certified to the highest internationally recognised safety standards for collaborative robotics. The care home deployed it in a space that violated the operational envelope specified in the manual. That is a deployment failure, not a manufacturing failure.*
- *Care home staff modified the robot's proximity sensor settings without authorisation. That modification voids the warranty and transfers liability entirely.*
- *If manufacturers are held liable for every way a complex system can be misused, no company will develop assistive robotics. The regulatory consequence of your position is that the technology disappears.*

The Care Home Director argues the manufacturer overstated the robot's capability for complex physical tasks.

- *The manufacturer's marketing material (the brochure, the demonstration video, the sales pitch) showed exactly this task being performed in a facility like ours. We deployed the product as marketed.*
- *There is a systematic gap between tested performance in controlled environments and real-world performance in a working care facility. The manufacturer knows this gap exists. They chose not to disclose it.*
- *We are responsible for the welfare of our residents. We relied on the manufacturer's assurances in good faith. If those assurances were misleading, the responsibility lies with the party that made them.*

KEY LEARNINGS Embodiment is not merely a technical property but a social one: a physical agent occupies space, exerts force, and triggers moral and legal responses that disembodied software does not. Moravec's paradox reminds us that our intuitions about what is "easy" or "hard" for intelligence are shaped by our embodied experience, and are regularly wrong. Kleist's dialogue makes visible the question that liability law must eventually answer: what work is the concept of "human" doing in law, and should physical presence, force, and risk be sufficient to extend it? The deployment of robots in care, policing, and domestic settings is a present policy challenge; the frameworks need to be built now, not after the first serious incident.

Teaser: when a machine has a body, acts with force, and answers to no one, what do we owe it?